

ANALYSING SPATIAL DATA FOR BIODIVERSITY CONSERVATION

JONÁŠ GAIGR

Nature Conservation Agency of the Czech Republic
Department of Biodiversity Monitoring



WHAT THIS SESSION COVERS

- **The data reality** – what Kosovo's published record shows, and why a thin baseline is no reason to wait.
- **Earth observation** – land-cover change and habitat fragmentation from Copernicus, and a species model built on them, available today at no cost.
- **One system, shared standards** – from project hard drives to a central biodiversity information system built on Darwin Core and INSPIRE.
- **Three working models** – Czech tiered open data, a Ukrainian GBIF viewer for impact assessment, English district level licensing.
- **Next steps** – what makes Kosovo's nature data fundable.

Kosovo* – this designation is without prejudice to positions on status, and is in line with UNSCR 1244/1999 and the ICJ Opinion on the Kosovo declaration of independence.



NO BASELINE IS NOT A REASON TO WAIT

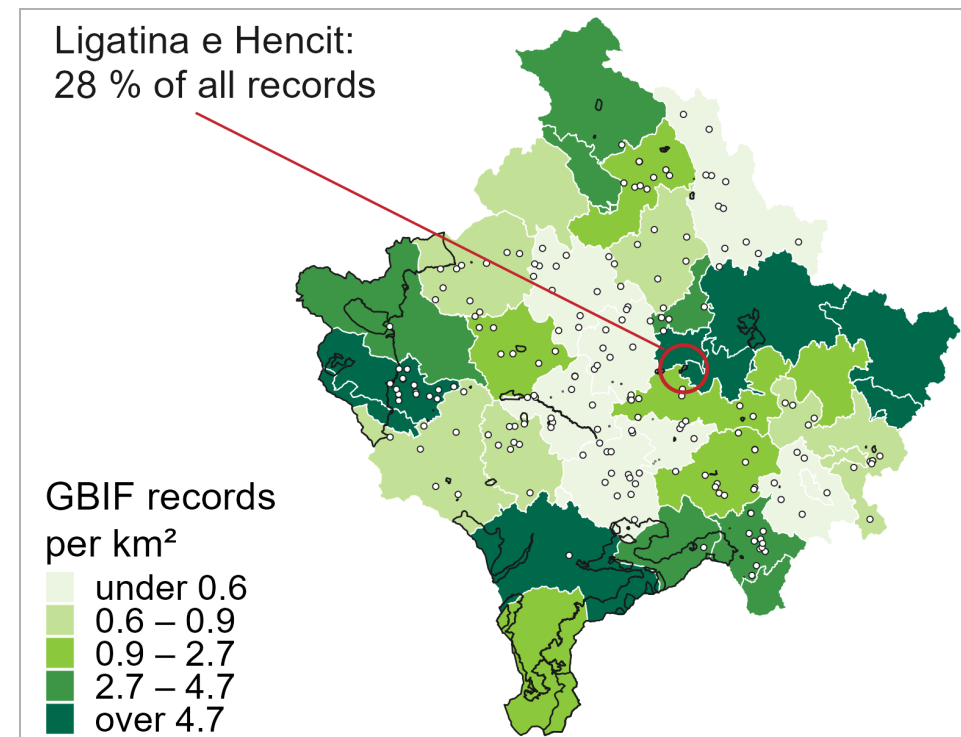
Permits, road alignments and quarry extensions are decided this week, with or without a map. Data does not add a decision – it makes the one being taken checkable.



KOSOVO'S PUBLISHED RECORD: THIN, AND CONCENTRATED

- **46,831** GBIF records – about **four per km²**.
- **28 %** of them come from **one wetland of 1.1 km²**.
- **48** protected sites have a boundary; **28** of those hold **no record**.
- **189** sites exist only as **a point** – nothing to intersect, report on or defend.

Published is not known: theses and survey reports hold far more, unseen.



Records per km² by municipality. Outlined: the 48 sites with a boundary. Circles: the 189 point-only sites. GBIF extract behind this repository's report; sites from the EEA register, v24, July 2026.

A MISSING BASELINE IS NOT A REASON FOR INACTION

Decisions are taken anyway

Every impact assessment and permit already answers the biodiversity question – from data, or from a site visit somebody remembers.

The choice is between an answer that can be checked and one that cannot.

“No record” is not “no species”

Under the Habitats Directive the burden sits with the project: it must show, beyond reasonable scientific doubt, that a site will not be harmed.

A blank map is a risk to the developer and to the ministry – not a clean bill of health.

The past can still be observed

Nobody can survey 2016 again. The satellites already did – at 10 m, every few days, since 2015.

The Earth-observation archive is the one baseline that can still be collected **retrospectively**.

Start with the evidence that already exists for the whole territory and costs nothing.



LAND-COVER CHANGE FROM COPERNICUS



Sentinel-2, 15 August 2016. Bjeshkët e Nemuna National Park (yellow line) above Pejë.



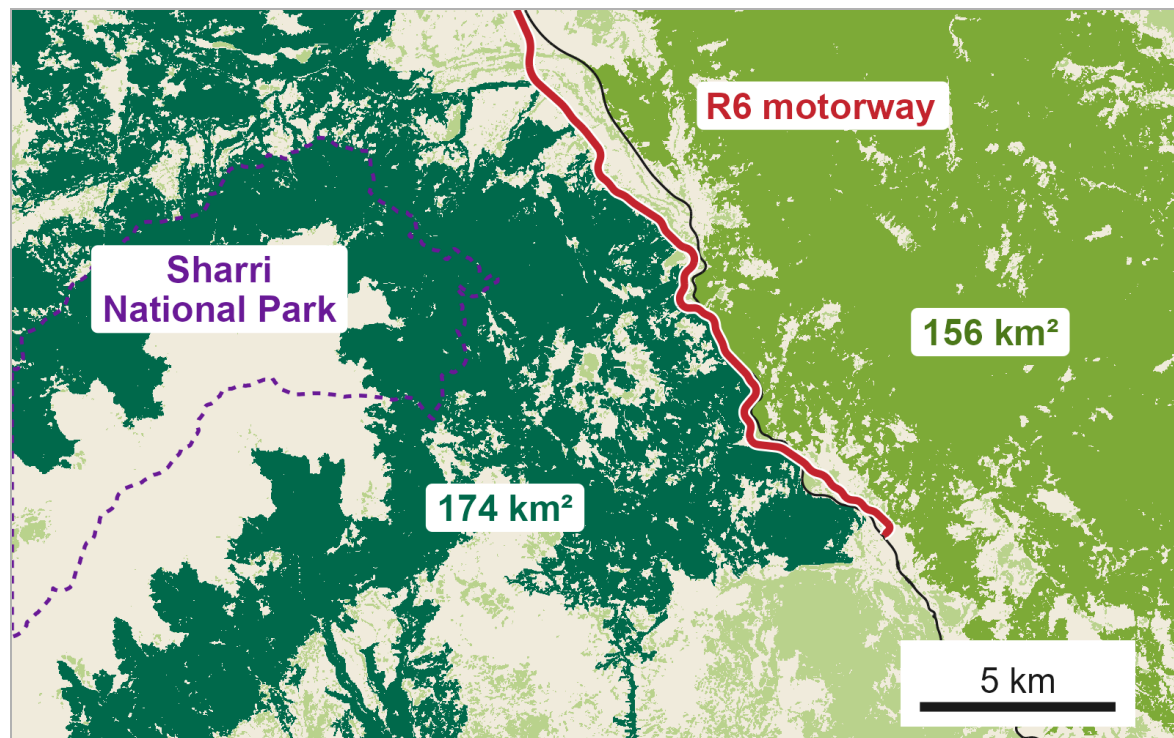
The same 5 km, 10 August 2025: a new road through the forest, inside the park.

Sentinel-2 – 10 m, free, archive from 2015. **CORINE Land Cover** – harmonised classes and change layers for 39 European countries, Kosovo included. Together: *where* land cover changed, *how much*, and *when*.

FRAGMENTATION: CONNECTION, NOT JUST COVER

- **Patch size and isolation** – how much habitat, in how many pieces, how far apart.
- **Effective mesh size** – the chance that two random points are still connected; the EEA's fragmentation measure.
- **Barriers** – motorways, fenced corridors, reservoirs, from the road and water network.

What it cannot see: species, habitat condition, anything under a canopy or in a cave. It tells field teams **where** to go – it does not replace them.

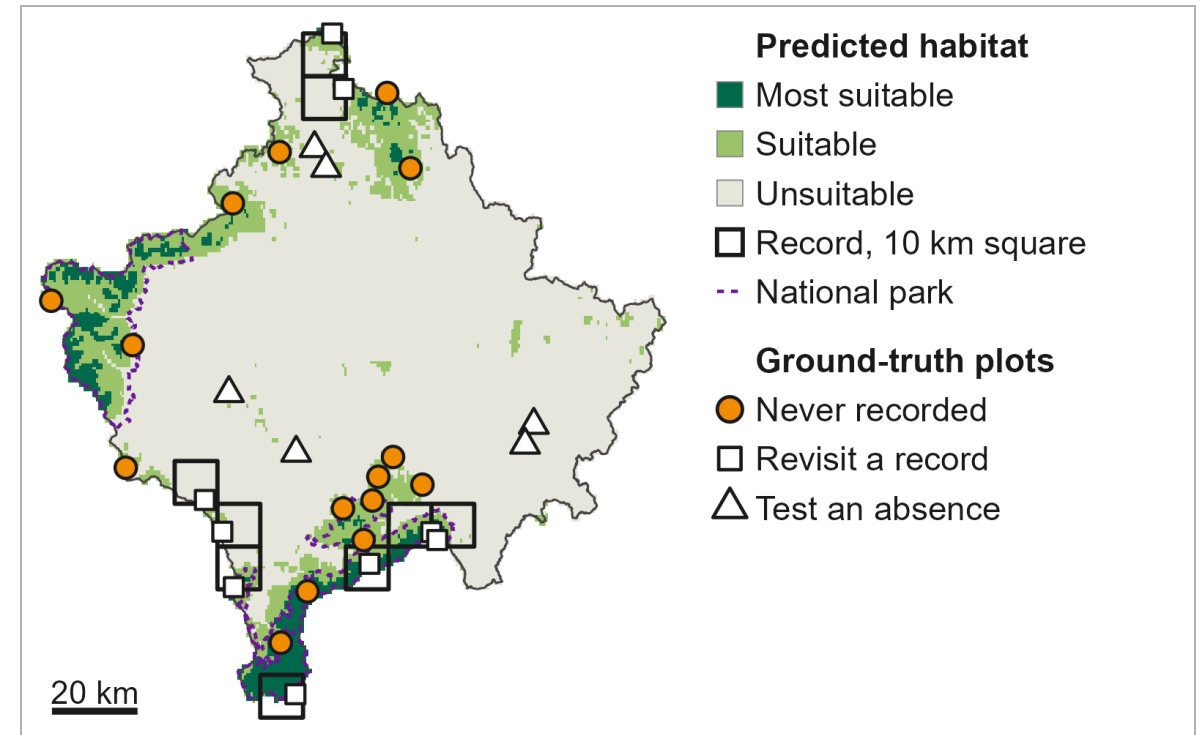


Forest, CLC+ Backbone 2021 (10 m), Kaçanik gorge. Each dark tone is one connected patch; pale, over 9,000 smaller ones. Black: the old main road.

SPECIES DISTRIBUTION MODEL: THE EROS BLUE

1. **Existing occurrence.** Kosovo: 25 records in nine 10 km atlas squares, one confirmed since 2007. The model learns instead from **31 precise records** in the same ranges, from six countries.
2. **Predicting occurrence.** WorldClim summer warmth and rainfall, CORINE 2018 alpine grassland, slope: **2,365 km²** suitable, over half of it in the two national parks. On regions held out: AUC 0.85.
3. **Validation by ground-truthing.** Never shown a Kosovo record, the model still finds habitat in **8 of the 9** atlas squares. Next: **30 plots** in one July–August flight season, absences recorded.

The trap: 545 of the 980 GBIF records under this name are another butterfly, *Aricia anteros*. Filter on the name the recorder wrote.



Relative habitat suitability for *Polyommatus eros*, 1 km, an ensemble of small models calibrated outside Kosovo. GBIF 10.15468/dl.265jgv, WorldClim 2.1, CORINE Land Cover 2018.

WHY PROJECT DATA DIES ON HARD DRIVES

The pattern – in every country, mine included

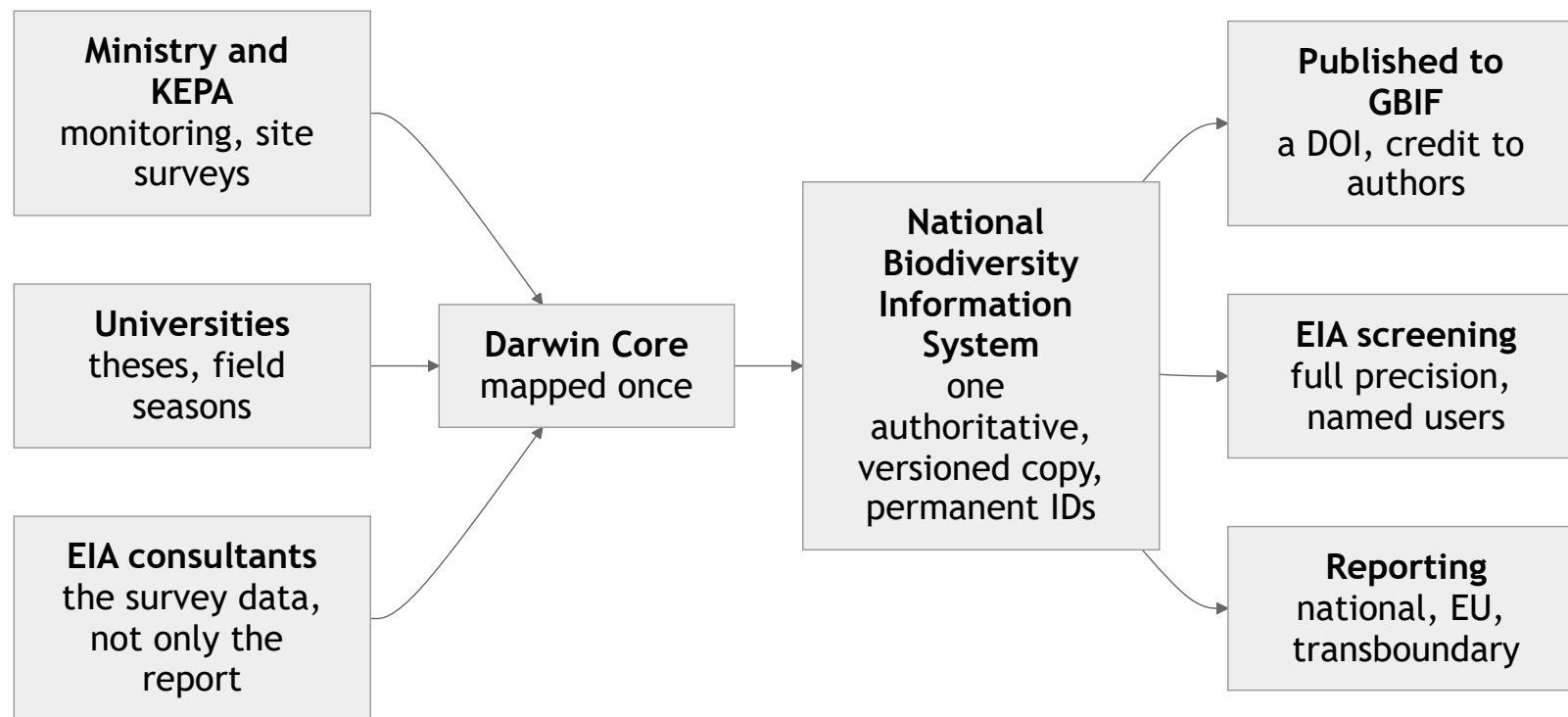
1. A project is funded; a consultant or a university team surveys.
2. The deliverable is **a report**. The data is an annex, or a file on a laptop.
3. The project closes. The laptop is replaced.
4. Five years later the same valley is surveyed again, because nobody could find the first survey.

Not a discipline problem – **a missing destination**.

What it costs

- **You pay twice**, with public money, for what you already knew.
- **You get no trend**. Two surveys that cannot be compared are not a time series – and trend is what reporting asks for.
- **You cannot defend a decision**. An assessment challenged in court needs the records, not a PDF conclusion.
- **Nobody gets credit**, so nobody hands anything over.

ONE CENTRAL BIODIVERSITY INFORMATION SYSTEM



INTEROPERABILITY: ONE RECORD, ACROSS BORDERS

Darwin Core – what a record *is*. The same terms in Pristina, Prague and Skopje, so records merge without a telephone call.

INSPIRE – how spatial data is *served*: *Protected sites, Species distribution, Habitats and biotopes*. The Kosovo Cadastral Agency’s geoportal has run INSPIRE-based services since 2013.

Across borders – Sharri and Bjeshkët e Nemuna continue into North Macedonia, Albania and Montenegro, which are building the Bern Convention’s **Emerald Network**. Shared standards let assessments meet at the border.

scientificName	<i>Bombina variegata variegata</i>
eventDate	2025-06-18T07:04
decimalLatitude	41.93105
decimalLongitude	20.69826
coordinateUncertaintyInMeters	200
basisOfRecord	HumanObservation
occurrenceStatus	present
license	CC BY-NC 4.0

One real Kosovo record in Darwin Core: yellow-bellied toad, Sharri National Park, June 2025 (GBIF occurrence 5897424625). Habitats Directive Annexes II and IV.

THREE SYSTEMS THAT WORK

A Czech database that is open without being reckless. A Ukrainian viewer that turns GBIF into an answer. An English licence that turned newt surveys into faster permits.



THE OPEN-DATA DILEMMA

The fear is legitimate

Orchids dug up, raptor nests robbed, tortoises taken, cave fauna collected. Precise public localities *have* been misused, in several countries.

Anyone who waves this away has not dealt with it.

The question is not *open or closed*. It is **who sees what, at what precision**.

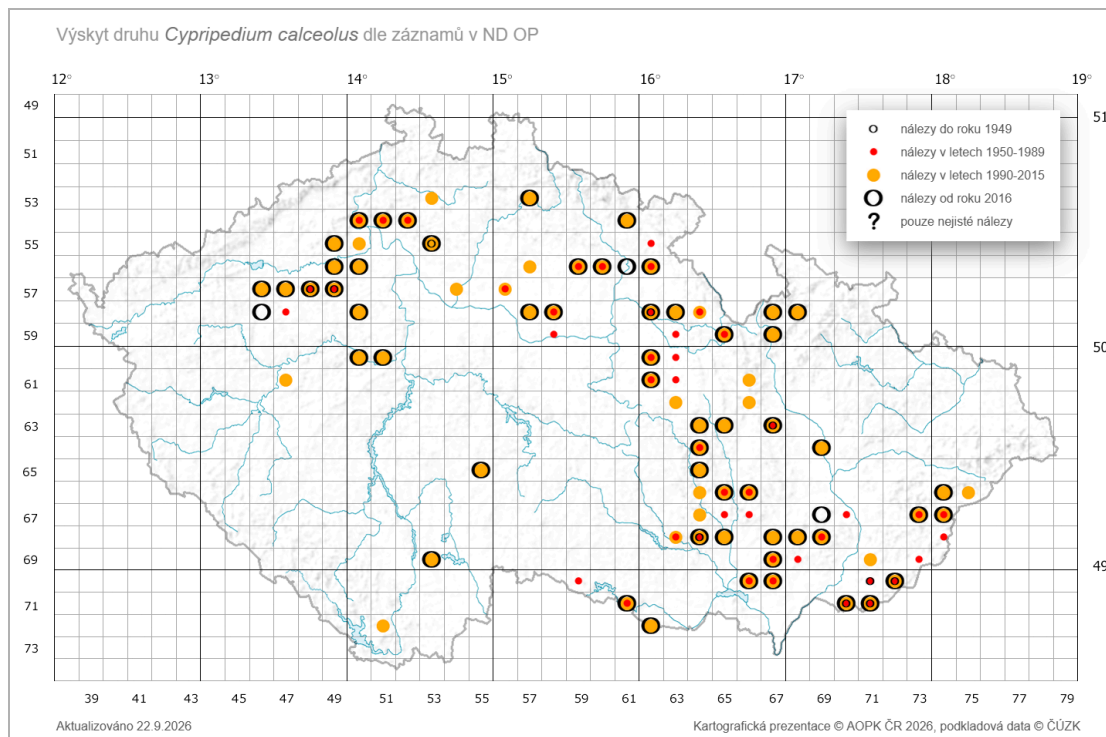
But closing the data protects the wrong people

- The collector **already knows the site**. The road engineer does not.
- To a planner, a site with no record is **an empty field**.
- Most sites are lost not to a poacher, but to **a bulldozer and a blank map** – lawfully.



NDOP: OPEN BY DEFAULT, PRECISE BY PERMISSION

- **42 million** published records – the great majority public.
- **Public view:** sensitive species are **generalised**, not hidden – the species visibly occurs in the area; the exact site does not show.
- **Professional view:** full precision for public administration and named users under a **data-use agreement**. Access is logged.
- **Governance, not software:** a named owner of the sensitive list, a review cycle, and species that can come **off** the list.



The public view of a sensitive species: present in the square, not pinpointed.

WHAT THE PLANNER SEES: THE FULL-PRECISION RECORD

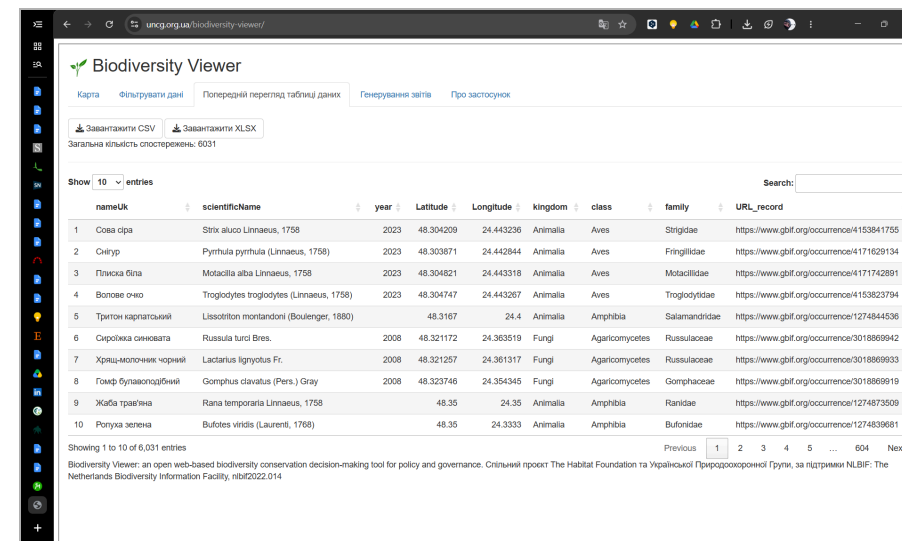
Nález ID	Mapa	Akce	Druh	Lokalizace	Datum od	Datum do
57380046	3 		 <i>Triturus cristatus</i> NEG 	Cihelna Kunín - hliniště, Lenčina tůň	20.7.2023	20.7.2023
57380044			 <i>Triturus cristatus</i> NEG 	Cihelna Kunín - pískovna, úzká	20.7.2023	20.7.2023
57380043	4 		 <i>Triturus cristatus</i> NEG 	Cihelna Kunín - Hliniště, Štěpánova tůň	16.5.2023	16.5.2023
57380042	3 		 <i>Triturus cristatus</i> NEG 	Cihelna Kunín - hliniště, Lenčina tůň	16.5.2023	16.5.2023
57380041	2 		 <i>Triturus cristatus</i> NEG 	Cihelna Kunín - hliniště, Nová tůň	16.5.2023	16.5.2023
57380040	2 		 <i>Triturus cristatus</i> NEG 	Cihelna Kunín - pískovna, úzká	16.5.2023	16.5.2023
57379976			 <i>Triturus cristatus</i> 	Cihelna Kunín - Hliniště, Štěpánova tůň	20.7.2023	20.7.2023

NDOP, logged-in view: great crested newt (*Triturus cristatus*) at one site. Pond-level localities, dates, photographs – and, marked **NEG**, recorded **absences**: ponds surveyed where the newt was *not* found. Absences are what make a trend possible. The same species is the subject of the English case at the end of this section.

RAW DATA IS NOT AN ANSWER

```
gbifID,scientificName,species,vernacularName,kingdom,phylum,class,
6500474327,"Mantis religiosa (Linne, 1758)",Mantis religiosa,Pr
6500462703,"Issoria lathonia (Linnaeus, 1758)",Issoria lathonia
6500450749,"Podarcis muralis (Laurenti, 1768)",Podarcis muralis
6500428815,Armeria alpina Willd.,Armeria alpina,Alpen-Grasnelke
6500295185,Gentiana acaulis L.,Gentiana acaulis,Gentiane De Koc
6500290475,Artemisia vulgaris L.,Artemisia vulgaris,Burot,Plant
... 46,825 more rows, 40 columns
```

What a developer receives: the Kosovo GBIF extract. Forty columns, scientific names, no legal meaning.



The screenshot shows the Biodiversity Viewer interface. At the top, there are tabs for 'Карта', 'Фильтрация данных', 'Попередний просмотр таблиц данных', 'Генерирование записей', and 'Про загрузку'. Below these, there are buttons for 'Загрузить CSV' and 'Загрузить XLSX'. A search bar is present on the right. The main table displays a list of species records with columns: nameUk, scientificName, year, Latitude, Longitude, kingdom, class, family, and URL_record. The table shows 10 entries, with a 'Showing 1 to 10 of 6,031 entries' message at the bottom. The footer contains a brief description of the Biodiversity Viewer and its funding sources.

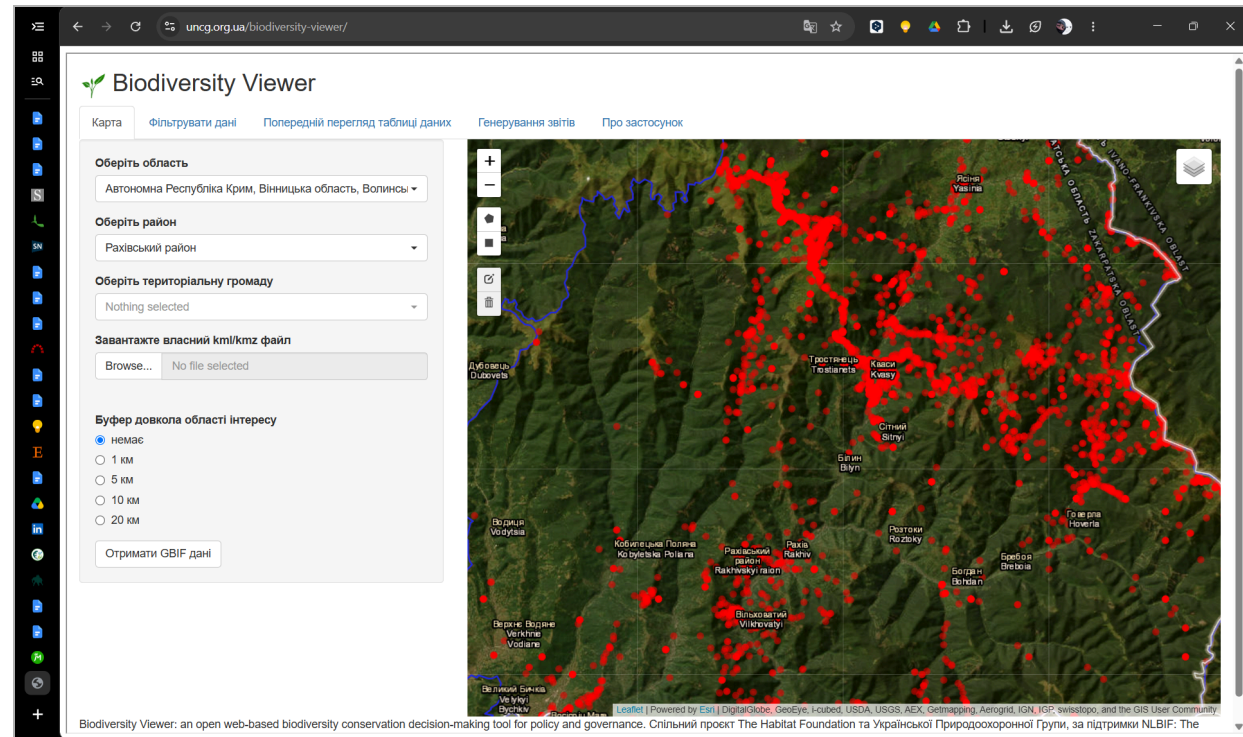
	nameUk	scientificName	year	Latitude	Longitude	kingdom	class	family	URL_record
1	Сова опа	Strix aluco Linnaeus, 1758	2023	48.304209	24.443236	Animalia	Aves	Strigidae	https://www.gbif.org/occurrence/4153841755
2	Синица	Pyrrhula pyrrhula (Linnaeus, 1758)	2023	48.303871	24.442844	Animalia	Aves	Fringillidae	https://www.gbif.org/occurrence/4171629134
3	Птица бела	Motacilla alba Linnaeus, 1758	2023	48.304821	24.443318	Animalia	Aves	Motacillidae	https://www.gbif.org/occurrence/4171742891
4	Волосок очко	Troglodytes troglodytes (Linnaeus, 1758)	2023	48.304747	24.443267	Animalia	Aves	Troglodytidae	https://www.gbif.org/occurrence/4153823794
5	Тритон карпатский	Lissotriton montandoni (Boulenger, 1880)		48.3167	24.4	Animalia	Amphibia	Salamandridae	https://www.gbif.org/occurrence/1274844536
6	Сиренка синеватая	Russula turci Bres.	2008	48.321172	24.383519	Fungi	Agaricomycetes	Russulaceae	https://www.gbif.org/occurrence/3018869942
7	Храп-молочник черный	Lactarius lignyotus Fr.	2008	48.321257	24.381317	Fungi	Agaricomycetes	Russulaceae	https://www.gbif.org/occurrence/3018869933
8	Гомфид булавовидный	Gomphus clavatus (Pers.) Gray	2008	48.323746	24.354345	Fungi	Agaricomycetes	Gomphaceae	https://www.gbif.org/occurrence/3018869919
9	Жаба трапезная	Rana temporaria Linnaeus, 1758		48.35	24.35	Animalia	Amphibia	Ranidae	https://www.gbif.org/occurrence/1274873509
10	Полуха зеленая	Bufo viridis (Laurenti, 1768)		48.35	24.3333	Animalia	Amphibia	Bufonidae	https://www.gbif.org/occurrence/1274839081

What a developer needs: the species in this area, and what the law says about each.

THE UNCG BIODIVERSITY VIEWER: GBIF, FILTERED BY LAW

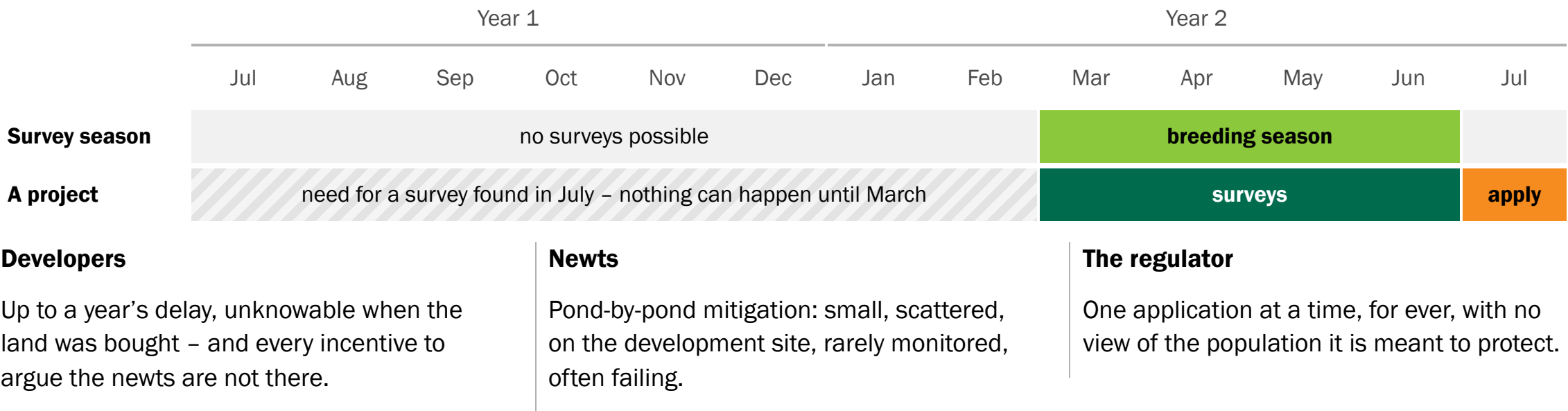
1. **Area** – district, drawn polygon or KML file, plus a buffer.
2. **Records** – the GBIF occurrences inside it.
3. **Law** – Red Data Book, Bern Convention, EU Habitats and Birds Directive annexes, IUCN, regional lists.
4. **Report** – a table and a DOCX, ready for the EIA file.

Open source (MIT). Built by UNCG and The Habitat Foundation, funded by NLBIF.



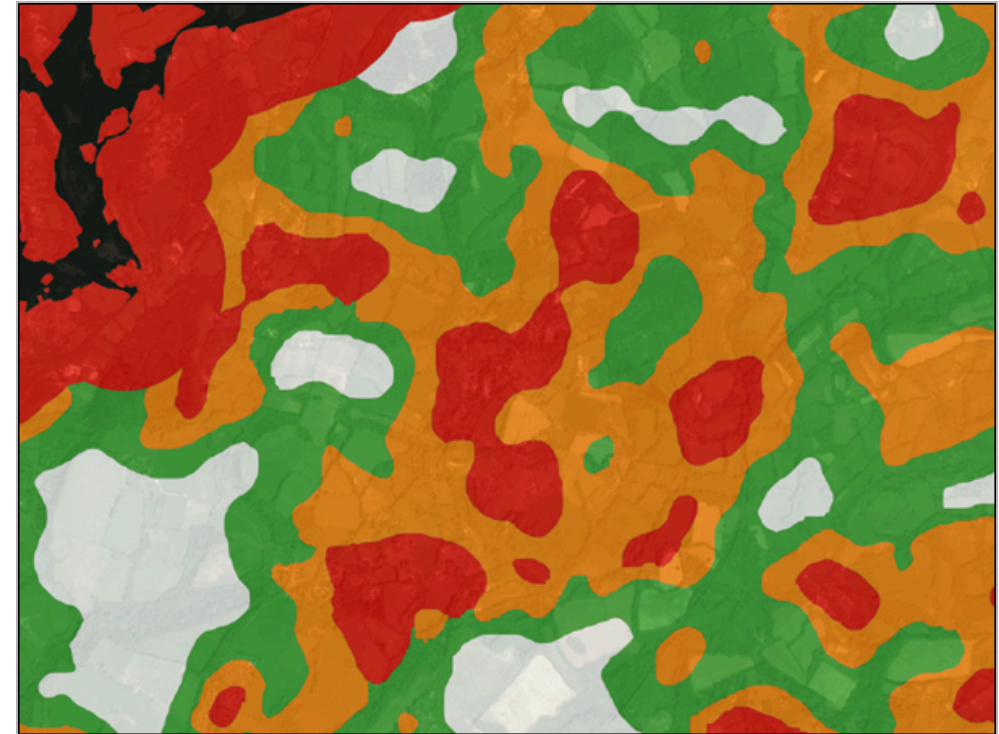
For Kosovo: replace one table – the national protected-species list – and the code is unchanged.

NEWT SURVEYS ON THE CRITICAL PATH



DISTRICT LEVEL LICENSING: MAP ONCE, PAY ONCE

- Newt presence **modelled across a whole district in advance**, from existing survey data and habitat.
- **Impact risk zones:** **RED** **AMBER** **GREEN** – the developer checks the zone instead of commissioning a seasonal survey.
- A **conservation payment**, set by the impact, replaces bespoke mitigation. The certificate goes in with the planning application.
- The money is pooled: **at least four ponds** created or restored for every occupied pond lost, managed for **25 years**.



Impact risk zones for one district. *[Source and licence to be recorded when the map is chosen.]*

NEXT STEPS: GOVERNANCE THAT ATTRACTS FUNDING

Govern – first 12 months

- Name **one owner** for biodiversity data.
- Adopt an **access and sensitive-species policy**.
- Draw boundaries for the **189 point-only sites**.
- Publish **three datasets** to GBIF in Darwin Core.
- Build the **Copernicus change baseline**.

Build – 1 to 3 years

- A **pilot** on two sites – Sharri and Ligatina e Hencit – to a draft SDF.
- A central **information system** with tiered access.
- An **area-screening tool** for EIA – the Ukrainian model, Kosovo's lists.
- **INSPIRE services** for protected sites, through the national geoportal.
- **Repeat monitoring**, with absences recorded.

Fund

- Funders back institutions that can show a **baseline, an owner and a data series** – not intentions.
- **IPA III** and the **Green Agenda for the Western Balkans** both prioritise nature.
- A **licensing charge** on the English model is domestic conservation money that grows with development.

Data governance turns Kosovo from a grant applicant into a credible partner.

SOURCES

- Biodiversity Information Standards (TDWG). n.d. *Darwin Core*. <https://dwc.tdwg.org/>.
- Council of Europe. 1979. *Convention on the Conservation of European Wildlife and Natural Habitats*. <https://www.coe.int/en/web/bern-convention/emerald-network>.
- Council of the European Communities. 1992. *Council Directive 92/43/EEC on the Conservation of Natural Habitats and of Wild Fauna and Flora*. <https://eur-lex.europa.eu/eli/dir/1992/43/oj>.
- European Commission, Directorate-General for Neighbourhood and Enlargement Negotiations. 2026. *TAIEX Expert Mission on Geographic Information Systems (GIS) and Biodiversity Data Management*. Case ID ETT IND/EXP 82606, Pristina, 28–30 September 2026.
- European Environment Agency. 2026. *Nationally Designated Areas (NatDA, Formerly CDDA)*. <https://doi.org/10.2909/028003e7-7585-4d69-92fc-7f81e0cc2340>.
- European Environment Agency. n.d. *CORINE Land Cover*. Copernicus Land Monitoring Service. Accessed 22 September 2026. <https://land.copernicus.eu/en/products/corine-land-cover>.
- European Environment Agency, and Swiss Federal Office for the Environment. 2011. *Landscape Fragmentation in Europe*. EEA Report No 2/2011. European Environment Agency. <https://www.eea.europa.eu/publications/landscape-fragmentation-in-europe>.
- European Parliament and Council of the European Union. 2007. *Directive 2007/2/EC Establishing an Infrastructure for Spatial Information in the European Community (INSPIRE)*. <https://eur-lex.europa.eu/eli/dir/2007/2/oj>.
- European Parliament and Council of the European Union. 2021. *Regulation (EU) 2021/1529 Establishing the Instrument for Pre-Accession Assistance (IPA III)*. <https://eur-lex.europa.eu/eli/reg/2021/1529/oj>.
- European Space Agency. n.d. *Sentinel-2 Mission Guide*. <https://sentiwiki.copernicus.eu/web/sentinel-2>.
- European Union, and European Space Agency. n.d. *Copernicus Data Space Ecosystem*. <https://dataspace.copernicus.eu/>.
- Fick, Stephen E., and Robert J. Hijmans. 2017. 'WorldClim 2: New 1-Km Spatial Resolution Climate Surfaces for Global Land Areas'. *International Journal of Climatology* 37 (12): 4302–15. <https://doi.org/10.1002/joc.5086>.
- Gaigr, Jonáš. 2026. *Biodiversity of Kosovo – GBIF Data Analysis*. Quarto website and reproducible R workflow. <https://jonasgaigr.github.io/kosovo-biodiversity-data-workshop-2026/>.
- Global Biodiversity Information Facility. 2026a. *GBIF Occurrence Download: Kosovo*. <https://www.gbif.org/occurrence/download/0003504-260903145123482>.
- Global Biodiversity Information Facility. 2026b. *GBIF Occurrence Download: *Polyommatus eros*, 18.5–24.5° E, 39.5–44.5° N*. <https://doi.org/10.15468/dl.265jgv>.
- Global Biodiversity Information Facility. n.d. *Integrated Publishing Toolkit (IPT)*. <https://www.gbif.org/ipt>.
- Leaders of the Western Balkans. 2020. *Sofia Declaration on the Green Agenda for the Western Balkans*.
- Meha, Murat, Joep Crompvoets, Muzafer Čaka, and Denis Pitarka. 2015. 'Implementing a National Spatial Data Infrastructure for a Modern Kosovo'. *FIG Working Week 2015*. http://www.fig.net/resources/proceedings/fig_proceedings/fig2015/papers/ts04d/TS04D_meha_crompvoets_et_al_7451.pdf.
- Natural England. n.d. *Great Crested Newts: District Level Licensing Schemes for Developers, Ecologists and Landowners*. Accessed 22 September 2026. <https://www.gov.uk/government/publications/great-crested-newts-district-level-licensing-schemes-for-developers>.
- Nature Conservation Agency of the Czech Republic. n.d. *Nálezová Databáze Ochrany Přírody (NDOP) – Species Occurrence Database*. Accessed 22 September 2026. <https://portal23.nature.cz/nd/>.
- NatureSpace Partnership. n.d. *Great Crested Newt District Licensing Scheme: FAQs*. Accessed 22 September 2026. <https://naturespaceuk.com/district-licensing/faqs/>.
- The Habitat Foundation, and Ukrainian Nature Conservation Group. n.d. *Biodiversity Viewer*. Open-source R Shiny application, MIT licence. Accessed 22 September 2026. https://github.com/ABiatov/gbif_shiny_onlineviewer.
- United Kingdom. 2017. *The Conservation of Habitats and Species Regulations 2017*. <https://www.legislation.gov.uk/ukxi/2017/1012/contents>.

THANK YOU FOR YOUR ATTENTION!

github.com/jonasgaigr/kosovo-biodiversity-data-workshop-2026



aopk.gov.cz

